

Ahmed AbouEleyoun

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Experience

Software Engineer	Trufla	2025–Present
• Extended core backend REST API with request defaults, field population, and CRUD limit enforcement • Built CLI as primary internal API interface, replacing Swagger UI for team's day-to-day API work • Extended CLI so engineers could automate API tasks in scripts and pipelines, not just interactively • Added multi-tenant auth and config to CLI, including credential stores, auto reauth, and 1Password secrets • Replaced fragile migration scripts with CLI deploy system for reliable config-as-code • Added CLI payload templating and variable substitution so same scripts work across tenants and environments • Maintained multi-tenant config-as-code repositories and shipped them through CLI deploy system		

Education

Cairo University – PgD in Computer Science, GPA: 3.6/4.0	2025
Mansoura University – BS in Surveying and Geographic Information Systems	2022

Projects

Functional Programming Language

- Designed a lazily-evaluated functional language with Hindley-Milner type system
- Implemented recursive-descent parser with operator precedence parsing, sections, and scoped local binds
- Built HM type inference system with unification, let-polymorphism, type schemes, and constraint solving
- Supported primitives declarations with typed external FFI, linking to native assembly stubs
- Developed G-machine IR interpreter as reference implementation for validating compiler semantics
- Created pattern matching compiler with exhaustive coverage checking and redundant pattern detection
- Generated x86_64 native binaries via NASM with runtime supporting lazy thunks and bump-allocated heap
- Integrated with external assembler and linker (NASM, mold) to produce statically linked executables
- Implemented rich diagnostic reporting with source span highlighting for syntax and type errors

Game Engine

- Built a cross-platform windowing and input layer with GLFW, event-driven callbacks, and monitor management
- Developed an OpenGL 4.6 rendering pipeline with shader compilation (GLSL/SPIRV)
- Added orthographic and perspective cameras with frustum culling and automatic uniform buffer updates
- Integrated FreeType and HarfBuzz for font rendering with a dynamic glyph atlas and SDF support
- Built a composition-based ECS with sparse-set component pools, entity versioning, and component queries
- Designed a resource system supporting PNG image loading via libpng and memory-mapped font loading
- Created a utility suite with transform, timer, logger, bitmask, event bus, and string hashing
- Modularized the entire engine using C++20 modules for clean dependency management and faster compile times

Development Tools Package Manager

- Built a CLI tool in Go for managing LSP servers, debuggers, linters, and formatters across multiple ecosystems
- Designed pluggable registry system supporting GitHub and local file sources for package discovery
- Implemented multi-ecosystem package installation supporting 12+ package types (cargo, npm, pypi, gem, etc.)
- Built platform-aware asset resolution using templated expressions and target triple matching
- Created atomic install/upgrade workflow with staging directories and rollback on failure
- Implemented symlink-based binary management with receipts for clean uninstall and tracking
- Added file-locking to prevent concurrent operations corrupting shared package state
- Developed registry update system with checksum verification for secure package metadata sync

Skills

Languages: Rust, C, C++, Go, Python, TypeScript

Tools and Technologies: Docker, Git, PostgreSQL, MySQL, MongoDB, Django, React, Shell Scripting, OpenGL